

HARSHIT GOYAL

Software & AI Engineer — B.Tech Electrical Engineering, IIT Delhi

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EDUCATION

Indian Institute of Technology Delhi

2022 – 2026

B.Tech, Electrical Engineering; CGPA: 7.513

EXPERIENCE

Software Development Engineer Intern

Jul 2026 – Present

Samsung R&D Institute

Delhi

- Training and fine-tuning Vision-Language Models (VLMs) to build a custom multimodal solution addressing an unmet product need.

Software Development Intern (EDA)

May 2025 – Jul 2025

Cadence Design Systems

Noida

- Drove Verilog RTL-to-Pre-CTS synthesis flows for multiple design blocks in Cadence Genus and Innovus, taking designs from elaboration and logic synthesis through floorplanning and placement.
- Analyzed setup/hold timing slack, area, power, and standard-cell utilization at each stage, comparing QoR across runs to surface bottlenecks and guide constraint and optimization tuning.
- Built a Python netlist-parsing tool extracting per-instance cell-utilization and QoR metrics from mapped netlists, and migrated legacy TCL automation to Python — cutting script execution time and replacing manual report inspection.

Technology Consulting Intern

Jun 2024 – Jul 2024

PwC India

Gurugram

- Digitized financial-service workflows for a PACS cooperative network serving 130M+ farmers and automated the underlying data processing in Python, cutting end-to-end processing time by 15%.

AI Research Intern

Dec 2023 – Jan 2024

Cape Peninsula University of Technology

Cape Town

- Researched AI-enabled smart-microgrid technologies and edge-deployment constraints, building a structured evaluation framework and authoring a 32-source technical report to guide adoption decisions.

PROJECTS

QuantAI – Quantitative Investment Research Platform

github.com/goyal-harshit/quant-analyzer

FastAPI · Next.js 14 · TypeScript · PostgreSQL/TimescaleDB · Redis · Celery · Docker

- Built a full-stack quantitative research platform for NSE/BSE equities with a multi-factor screener and an event-driven backtester over TimescaleDB time-series storage.
- Integrated a RAG-grounded AI assistant with a swappable LLM backend (Ollama, OpenAI, Anthropic, Gemini, Groq) to answer research queries grounded in market and fundamentals data.
- Offloaded ingestion and backtest jobs to Celery/Redis workers; containerized with Docker and shipped GitHub Actions CI/CD.
- Hardened for production: JWT auth in httpOnly cookies with CSRF, per-source circuit breakers and rate limiters with fallback chains, Prometheus metrics, and structured JSON request logging.

Codebase Intelligence Platform (tree-sitter, ArcadeDB, ChromaDB, FastAPI)

github.com/goyal-harshit/codebase-intelligence-platform

- Ingests any Git repository, parsing source with tree-sitter into a language-agnostic code knowledge graph (ArcadeDB) plus a semantic vector index (ChromaDB) for structural and natural-language queries.
- Detects god objects, dead code, and circular dependencies via call/import graph traversal; parsed a 24-file Flask codebase in 0.28s, extracting 388 functions and 53 classes.
- Answers plain-English code questions through hybrid retrieval — natural language compiled to Cypher for structural queries with semantic vector fallback — and keeps large repos current via SHA-256 incremental re-indexing that re-parses only changed files.

Intuitive Prompt Engine – Gesture-Driven Image Generation

github.com/goyal-harshit/intuitive-prompt-engine

Python · MediaPipe · FastAPI · WebSocket · Ollama · FLUX/SDXL

- Generates images from webcam hand gestures alone, using MediaPipe hand tracking to map gestures to prompt actions with no keyboard input.
- Routes gestures through a swappable LLM-backed prompting layer (Ollama) into structured prompts, streaming generation over WebSockets to pluggable diffusion backends (FLUX/SDXL) behind FastAPI.
- Designed each pipeline stage behind an interface — frame source, landmark extractor, intent model, prompt generator, image backend — so any stage swaps without touching the rest; shipped with CI, Docker, and an MIT license.

AI/ML Fantasy Cricket Predictor – 9th Nationally, FIFS Gameathon 2025

Google Cloud & Dream11

- Built an automated CricSheet pipeline with match-wise feature engineering, then trained a Random Forest / Gradient Boosting / PyTorch player-embedding ensemble to model player performance — placing 9th nationally.
- Modeled player form, matchup, and venue effects, then selected final teams under budget and role constraints via constrained optimization over predicted scores.

TECHNICAL SKILLS & ACHIEVEMENTS

Languages: Python, C++, TypeScript, SQL, TCL

AI/ML: PyTorch, TensorFlow, scikit-learn, RAG, LLM APIs (OpenAI, Anthropic, Gemini, Groq), Ollama

Backend & Web: FastAPI, Next.js, Celery, Redis, WebSockets

Data & DevOps: PostgreSQL/TimescaleDB, ChromaDB, Docker, GitHub Actions, Git

Achievements: JEE Advanced 2022 – AIR 530 (GE) of 250,000+; JEE Main 2022 – 99.74 percentile of 1,000,000+